MedCheck

HEALTH VERIFICATION

AI-Powered Drug Interaction Analysis

Jack Klopson (@jklopson) & Jack Martin (@martin-jack22)

The Problem We Wanted to Solve

- According to the FDA, the risk of adverse drug reactions increases ***exponentially*** after a patient is on 4 or more medications

- Consumers need an accurate, easy way to find the information that is essential to their health

- IBM Granite AI is perfect for use in tackling this critical issue

Our Solution Overview -

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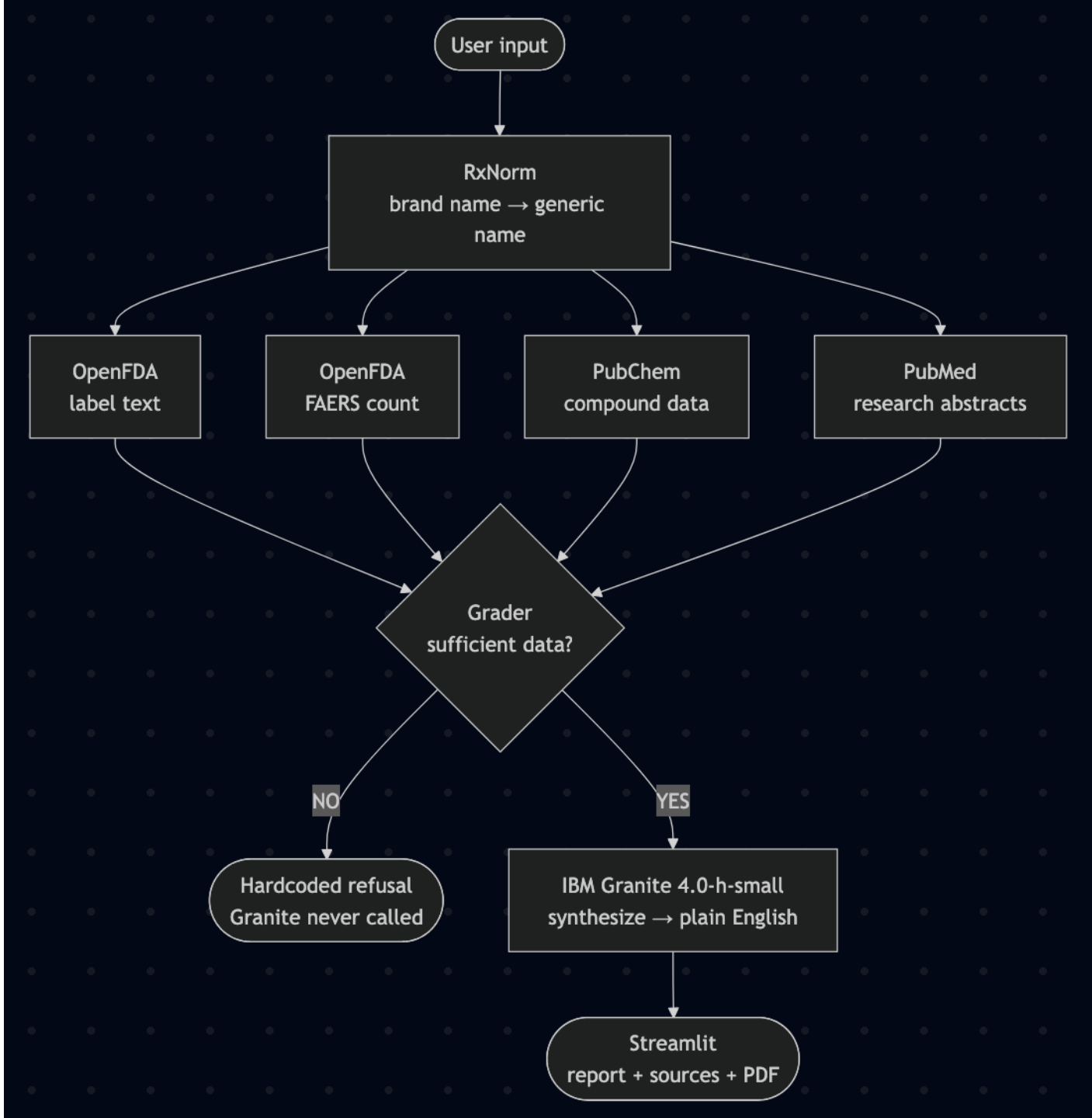
- Takes a list of medications and reasons through their interactions step by step
- Returns a clear, easily downloadable summary of risks and severity
- Powered by IBM Granite AI, real FDA drug data, and PubMed/PubChem

Tech Stack

- IBM Granite
- Python
- RxNorm
  - NIH drug name database
- OpenFDA
  - FDA regulated drug interaction data, and adverse event counts (FAERS)
- PubChem
  - NIH compound database
- PubMed
  - NCBI research database: peer-reviewed abstracts on drug pair interactions
- Streamlit
  - Python web UI

System Architecture / Agent Flow



Live Demo!

Challenges, Lessons Learned, and Future Work

- Challenges
  - We had some struggles getting the Streamlit to work and getting the returned information from the model to be properly entered into the downloadable PDF
  - Reliable data retrieval and synthesis
- Learning
  - Planning ahead on merge points as a team to avoid conflicts
  - Developing iteratively to ensure we had a working project to show
- Future work could include:
  - Expanding and optimizing the data retrieval
  - Linking valid sources from the queried databases
  - Developing better metric for 'sufficient data' about a given pair of drugs (alongside ReAct agent instead of ToolRAG)

GitHub Repo

- GitHub repo: https://github.com/jklopson/medication-interaction-agent